Project Design Phase-II
Technology Stack (Architecture & Stack)

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Team ID	EN25CS3011097
Project Name	Global Olympic Sports Performance Analysis

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	SQL (Snowflake), Tableau calculated fields	SQL verifies totals; calculated field creates Total Medals
2	Database / Data Warehouse	Warehouse used to store and manage structured business data for analysis and reporting.	Snowflake / CSV	Holds 8,988 records in one table with a record-type column
3	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stage / COPY INTO / Tableau Data Connections	Loads athelete_events.csv and connects it to Tableau
4	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Tableau Desktop, Snowflake SQL	Filters by dataset_type, COUNTD of athletes, total medals
5	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau (bar and stacked bar charts)	Seven worksheets, dashboard and story
6	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public / Tableau Desktop	Interactive filters and story navigation for end users